

Find Us — Clean Research Synthesis (RESEARCH_DUMP.md)

Project: Find Us — Infrastructure-Free Cooperative Spatial Navigation for Emergency Crowds **Document:** Deliverable 1 (clean research synthesis; the authoritative clean view) **Date:** 2026-09-23 **Companions:** PAPER.md (paper draft), SOURCES.md (source registry), GAPS.md (gap registry), [website/](research/website/) (static research/interview site), README.md (navigation).

0. Integrity Preamble (read first)

This document is written under the project's honesty constitution. Every research claim is tagged with exactly one of:

Tag	Meaning
[RES]	Research decision / design claim — defensible as architecture, publishing-safe as design.
[RES:MATH]	Mathematical fact (e.g., SE(2) adjoint composition on a Lie group) — provable, publishable as math.
[EXP:NOT-RUN]	Designed/derived but not executed . No simulation run artifact, no hardware run. Explicitly NOT a result.
[EXP:NOT-RUN] [SIM:EST]	A number that <i>looks</i> like a measured benchmark but is a design-parameter estimate from the wire/model. Retagged from a phantom "measured" line. See §1.4.
[RES:NOT-VERIFIED]	Claim known to be plausible but not resolved by this research pass.
[PRIOR]	Prior art (external) — with verification status, not invented.
[SPEC]	Speculation — clearly separated, never presented as result.

Rules applied in this document:

1. No fabricated citations. Every external source is in SOURCES.md with a [VERIFIED] / [PARTIAL] / [UNVERIFIED] status.
 2. No fabricated results. Any number presented "as measured" that has no run artifact is retagged [EXP:NOT-RUN] [SIM:EST] or removed.
 3. No claimed prior art. Anything not confirmed by an actual retrieved source is excluded or flagged.
 4. No "seamless" language. Graceful degradation is a **design target** ([RES]), not a demonstrated property.
 5. GPS is a weak skeleton, never the navigation source.
 6. UNKNOWN is a valid first-class state, never an error.
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1. Executive Summary / Claim Map

1.1 The problem

A crowd of heterogeneous smartphones (different sensors, calibration, coordinate frames, noise, mobility, connectivity) must estimate each other's positions **relative to one another** with **no permanent infrastructure** and **no reliable absolute reference** (especially in emergencies: indoor, urban canyon, GPS-denied, crushed crowds).

Core insight (this project): GPS is NOT the navigation source. GPS is a coarse global **skeleton** — a weak constraint that (a) fixes the global gauge/rotation/translation ambiguity, (b) seeds the optimizer, (c) anchors the cooperative frame, (d) provides weak calibration priors, (e) enables gross drift detection. **Precision comes from peer observations**, not GPS.

1.2 The central claim (thesis)

A heterogeneous, mobile, infrastructure-less crowd can maintain a coherent, honest, *mode-laddered* relative spatial representation — where the mode (metric → differential → topological → historical → ghost) explicitly tags every observation — **without** requiring a shared absolute frame, continuous sigma, or a permanent anchor, **and** can detect its own failure states as first-class "I don't know" conditions.

1.3 Claim-level status (14 claims, honest)

#	Claim	Status	Today
C1	Relative frames (SE(2)/SE(3)) without absolute reference	[RES:MATH]	Architecture — publishable
C2	SE(2) relative transforms compose pairwise	[RES:MATH]	Math fact — publishable
C3	Wire budget fits BLE advertising ceilings	[RES] + [EXP:NOT-RUN]	Design — untested on real BLE
C4	RSSI log-distance gives metric range	[EXP:NOT-RUN]	Hypothesis — needs calibration
C5	Torso-scan / two-walk gives bearing	[EXP:NOT-RUN]	Hypothesis — unverified
C6	2-hop metric → topological ladder	[RES]	Design — coherent, untested
C7	Discrete confidence classes (CERTAIN...GHOST) replace continuous σ	[RES]	Design — strong, publishable
C8	Ghost decay: nothing expires, everything decays	[RES]	Design — strong, publishable
C9	Mode transitions are explicit events	[RES]	Design — strong, publishable
C10	Symmetric reciprocal-consistency (triangle) check	[RES]	Design — strong, publishable
C11	Hop count multiplies confidence	[RES]	Design — cheap, publishable
C12	ANCHOR/DRIFTER role tagging	[RES]	Design — strong
C13	UNKNOWN as first-class, non-error state	[RES]	Design — strong
C14	"Graces gracefully" / "74.8% at 2,500 nodes"	[RES] + [EXP:NOT-RUN]	Aspirational — NOT a result

Key rule: [RES]/[RES:MATH]/[RES:NOT-VERIFIED] claims can go into a paper as *design*. [EXP:NOT-RUN] claims may appear ONLY as future work, never as a result.

1.4 THE retag (integrity-critical correction)

The research log previously carried one line written as a measured benchmark with **no run artifact** and **no** [EXP] marker — a phantom:

"Bystander battery drain is reduced from \$85.3\text{ mAh/day}\$ down to \$1.39\text{ mAh/day}\$... a \$61.8\times\$... Confidentiality score remains \$1.00\$."

Verification (this session): the referenced simulation `sim_encrypted_relay_overlay.py` does **not exist** in the repository (searched: 0 hits); no run artifact exists; the [EXP:NOT-RUN] marker count for the H6 benchmarking line has been **re-verified = 2, confirmed**. The line is now retagged in `research/research-log.md`:48 to:

[EXP:NOT-RUN][SIM:EST] – no run artifact exists for H6 benchmarking; the numbers below are DESIGN-PARAMETER ESTIMATES from the wire/model (31-byte frame = 8-byte Public Envelope + 19-byte Authenticated Private Core), NOT measured benchmark results.

Why this matters: a hostile reviewer who greps for `sim_encrypted_relay_overlay.py` (or any executed-run registry) will find nothing. The system's own "confidentiality financial honesty" constitution forbids presenting an

estimate as measured. **This one line, if left, would classify the whole paper as fabricated.** It is now honest.

2. Problem Decomposition (research questions, grounded)

2.1 The seven-layer fallback hierarchy

Level	Mode	Description
0	GPS skeleton + cooperative	Global frame + precise relative positions
1	Cooperative only (GPS-denied)	Relative, internally consistent, no global anchor
2	Cooperative + partial peers	Degraded precision, topological still useful
3	Dead-reckoning only	Approximate, drift accumulates
4	UNKNOWN	Honest "I don't know" — never fabricated

Design target (not result): the fallback hierarchy must function at Level 1 indefinitely; enjoy, degrade gracefully through 2–3, and report Level 4 honestly. **[RES]**

2.2 The mode ladder (the spatial model's backbone)

METRIC (hops 1–2) → measured vectors, SE(2) relative transforms, torso/two-walk bearing
DIFFERENTIAL (any) → Direction-of-Improvement (gradient, two-walk, no frame, no spin)
TOPOLOGICAL (hops 3+) → hop count only, no coordinates, no bearing claim
HISTORICAL (beyond) → last-known with age
GHOST (never expires) → "was there once," decays further, never deleted

No mode masquerades as another. Every observation is tagged with its mode. Every mode transition is an explicit event.

2.3 Research questions driving the paper (RQ1–RQ15, condensed)

- RQ1 (wire budget) — max usable per-payload bytes under 31-byte / 27-byte / 23-byte / iOS ceilings. **[RES; platform ceilings derived, hardware re-verify [EXP:NOT-RUN]]**
- RQ3 (spoof/confidentiality) — does a partitioned frame with ECC+MAC detect spoof/replay? **[RES: unit-tested protocol [EXP]]; field [EXP:NOT-RUN]**
- RQ11 (SOS gradient bearing) — does coupled SOS-gradient + VRLG give navigable bearing $\leq 20^\circ$? **[RES: simulated [EXP]]; field [EXP:NOT-RUN]**
- RQ13 (baro floor-gating) — multi-floor routing without GPS? **[PRO; baro models only [EXP:NOT-RUN]]**
- RQ15 (fallback selection) — which fallback under which environment? **[PRO; platform-gated [EXP:NOT-RUN]]**

3. Layer-by-Layer Research Synthesis (the clean dump)

LAYER 1 — GPS SKELETON LAYER **[RES]**

Problem: GPS is coarse, intermittently available, and unreliable indoors/urban canopy. It must be a weak prior, not navigation.

Findings:

- **L1.1 (GPS as weak constraint):** Treat GPS as a coarse gauge-fixing / calibration prior in a factor graph, not a hard constraint. Use robust loss (Huber) with inflated covariance. **[RES]**
- **L1.2 (availability transition):** Smooth GPS-available $\leftrightarrow$ GPS-denied transitions; no discontinuity. Detect mode via availability + quality + consistency. **[RES]**
- **L1.3 (GPS for calibration):** Weak global constraint for step-length / compass calibration when briefly available. **[RES]**

- **L1.4 (GPS-denied bootstrapping):** Anchor-free bootstrap using gravity, magnetic heading, leader election, relative frame conventions. [RES]

Research problems (open): What minimal information must GPS contribute? What is a "weak prior" formulation with honest uncertainty? [EXP:NOT-RUN]

LAYER 2 — COOPERATIVE ESTIMATION LAYER [RES] + [EXP:NOT-RUN]

Problem: Phones observe each other (RSSI, UWB ranging, bearing, IMU-derived motion), but each has an independent frame, sensor, calibration, and noise model.

Findings:

- **L2.1 (peer observation modalities):** Surveyed. RSSI ranging (log-distance, noisy, needs calibration), UWB (accurate, sparse hardware), bearing (torso scan / two-walk), dead reckoning (drifts). [RES][PRIOR:VERIFIED] — see SOURCES.md.
- **L2.2 (factor graph formulation):** Cooperative localization as a factor graph / graph SLAM over relative measurements; uncertainty propagates through SE(2) adjoint composition. `core/relationships.py` implements composition + cycle checks. [RES:MATH]
- **L2.3 (distributed vs centralized):** Distributed message-passing (trickle, trickle-absorption) suppresses broadcast storms; local triangle consistency catches errors in $O(1)/\text{edge}$. [RES]
- **L2.4 (frame alignment):** The hardest problem. Two-walk Direction-of-Improvement sidesteps frame alignment entirely at the local tier. [RES]

The two fundamental fixes (contribution candidates):

1. **Discrete confidence classes replace continuous σ :** CERTAIN / LIKELY / GUESSED / UNKNOWN — two bits on the wire, honest, zero propagation math, recomputed locally at receiver. [RES]
 2. **Ghost decay replaces expiry:** FRESH $\rightarrow$ DEGRADED $\rightarrow$ HISTORICAL $\rightarrow$ GHOST; nothing deleted, everything loses authority with age/hop distance but never becomes "unknown-fabricated." [RES]
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LAYER 3 — CALIBRATION LAYER [RES] + [EXP:NOT-RUN]

Problem: Different phones, different calibration; errors accumulate and compound through relays.

Findings:

- **L3.1 (self-calibration):** Gyro bias (static), accel bias (gravity), mag hard/soft iron (rotation), step length (GPS/peer). [RES][PRIOR:PARTIAL]
- **L3.2 (peer-validated calibration — CORE NOVELTY):** When A and B observe each other reciprocally, inconsistent results (A says B is 10 m away, B says A is 8 m away, both confident) flag calibration error **without ground truth**. Reciprocal inconsistency becomes the calibration oracle. [RES:MATH] formulation; [EXP:NOT-RUN] validation
- **L3.3 (recalibration triggers):** Innovation/consistency thresholds, drift detection, context change. [RES]
- **L3.4 (calibration state):** Continuous validity, uncertainty, temporal decay, context-dependence. [RES]

Honesty note: L3.2 is the strongest novel claim but is **formulated only**; it requires the falsification experiments in GAPS.md/E1–E3 to move from [RES] to [EXP:RUN]. [RES:NOT-VERIFIED]

LAYER 4 — UNCERTAINTY & DEGRADATION LAYER [RES]

Problem: Know what you know, know what you don't, degrade honestly.

Findings:

- **L4.1 (uncertainty propagation):** SE(2) adjoint covariance propagation within the metric tier. [RES:MATH]
- **L4.2 (graceful degradation):** The mode ladder IS the degradation: metric $\rightarrow$ differential $\rightarrow$ topological $\rightarrow$ historical $\rightarrow$ ghost, each with defined failure state and UI rendering. [RES]
- **L4.3 (knowing when you don't know):** UNKNOWN is first-class. Every "I don't know" is renderable, never an error, never filled with fabricated data. [RES]

Degradation tiers (honest render):

- **PRECISE** (metric, <1 m) — hop ≤2, recent, direct
- **APPROXIMATE** (metric, >1 m) — hop ≤2, older
- **TOPOLOGICAL** (hop count) — hop 3+, no coordinates
- **HISTORICAL** (last known, aged) — beyond reach
- **UNKNOWN** (no observation) — valid state, "searching..."

[RES] — all design; **none measured**.

LAYER 5 — COMMUNICATION LAYER [RES] + [EXP:NOT-RUN]

Problem: Spatial observations must travel through byte-capped, lossy, partial-connectivity BLE meshes without leaking plaintext.

Findings:

- **L5.1 (measurement context):** Each observation must carry timestamp, frame ID, modality, uncertainty class, calibration state. Protocol designed. [RES]
- **L5.2 (partial connectivity / partition healing):** Reconciling independent spatial states on reconnection; stale-info handling; delay-tolerant / store-and-forward (/ data mule). [RES]
- **L5.3 (enough information):** Minimum sufficient exchange — not full state, not full covariance; discrete classes, hop counts, ghost decay, direction-of-improvement. [RES]

Wire-integrity findings:

- Trickle (RFC 6206) + absorption suppresses re-broadcast storms. [RES][PRIOR:VERIFIED]
 - Legacy BLE 31-byte ceiling partitions: Public Envelope + Authenticated Private Core. [RES] — **but benchmark numbers are** [EXP:NOT-RUN] [SIM:EST] **per §1.4.**
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LAYER 6 — APPLICATION LAYER [RES] + partial [EXP]

Findings:

- Android/iOS app architecture (sensor logging, BLE discovery/exchange, estimation, UI) designed; app/, android/, ios/, core/ implemented. [RES]
 - CI state: Android empty-message battery case + 5 iOS compile errors pending green (see README.md/CI). [EXP:NOT-RUN]
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4. The Core Novel Contributions (candidate, honest)

1. **The mode-laddered spatial model** — four/five explicit modes tagged at observation level, no mode masquerading as another, explicit mode transitions as events, ghost-decay instead of expiry. [RES] — the strongest defensible contribution.
2. **Discrete confidence classes** replacing continuous σ — "continuous sigma is false precision," honest classes, two bits, zero propagation math. [RES]
3. **Direction-of-Improvement** (differential, two-walk, gradient) replacing absolute bearing — sidesteps frame alignment. [RES]
4. **Peer-validated self-calibration** (reciprocal inconsistency as oracle) — novel, **formulated only**, requires experiments. [RES:NOT-VERIFIED]
5. **Symmetric composition invariant / triangle cycle check** — local $O(1)$ error detection, no global consistency engine. [RES:MATH]
6. **UNKNOWN as first-class, anti-fabrication failure state** — emergency systems must not fabricate. [RES]

Each requires prior-art confirmation (see §7).

5. What We CanNOT Claim Today (honest negative registry)

- "Gracefully degrades" as measured — design target only. [RES→[EXP:NOT-RUN]
- Any RSSI/bearing/ σ accuracy number — constants unmeasured. [EXP:NOT-RUN]
- 85.3→1.39 mAh/day, 61.8 $\times$, 1.00 — **phantom, retagged**. [EXP:NOT-RUN][SIM:EST]
- "74.8% at 2,500 nodes" as measured — simulated-design estimate. [SET: on-line]
- "2,000-node scale" — E31 explicitly [EXP:NOT-RUN].
- Any battery/CPU/confidentiality number — no run artifact, no real hardware.

6. Cross-Project Prior-Art Comparison (this study vs. the field)

Work	What it does	What it does NOT do (our relation)
CoCoA (Rangoni? per SOURCES)	Mobile anchor beacon + RSSI Bayesian localization w/ energy coordination	no mode-ladder, no ghost-decay, no peer-validated calibration
SPAWN (Wymeersch et al.)	Cooperative UWB localization, Factor-Graph/Message-Passing	metric-only; no degradation modes; privacy not first-class
NBP hierarchical-graph (Cui/Chen)	cooperative WSN localization with NBP layers	metric-only, error-mitigation focus; not ad hoc emergency crowd
OEPB (IETF I-D)	Emergency P2P BLE broadcast mesh	no spatial estimation integration
IgniRelay/DisasterMesh	BLE advertising mesh relays, store-and-forward	relay-only, no cooperative estimation
CrisisConnect/BlueSOS	BLE mesh SOS	no relative spatial estimation
CORELS/CLIPS	relative-coordinate cooperative localization	no emergency mode-ladder / ghost /
Patent US11627453	Emergency comms over non-persistent P2P BLE	mechanism TBD vs our relay overlay

[Basic: prior-art gap audit: full audit at [research/find-us-audit/01_prior_art_gap_audit.md](#); confirmatory search this session found no system combining mode-ladder + ghost-decay + discrete classes + peer-validated calibration.]

7. The Gap That Survives (the integration gap) [RES:NOT-VERIFIED — needs confirmatory search]

The single defensible research gap: No prior work integrates:

1. a **mode-laddered spatial degradation model** (metric/differential/topological/historical/ghost),
2. **discrete confidence classes** with explicit **ghost-decay** (never-expire),
3. **peer-validated reciprocal calibration** as the calibration oracle,
4. first-class **UNKNOWN/anti-fabrication** states,
5. all within a **byte-capped legacy-BLE relay mesh** for emergency crowds.

This is a **claims-level gap**, not a component gap: each component exists in isolation (SPAWN, CoCoA, DTN, Trickle, ghost gradients in robotics), but the **integration is unclaimed**. [PRIOR:PARTIAL] — **confirmatory one broad search done this session; no direct integration found; further per-layer RRQ searches are marked future work.**

8. Honest "State of the Art" (with error-flagged honesty)

- Relative-coordinate cooperative localization: **exists, substantial** (SPAWN, CoCoA, CLIPS, NBP). They are metric/normal-estimation systems.
- Emergency P2P BLE relay meshes: **exist** (OEPB, IgniRelay, DisasterMesh, BlueSOS, Vajra, patent).
- Trickle/broadcast suppression: **exists** (RFC 6206), applied in SOS meshes.

- Uncertainty in WSN coop localization: **exists** (CRLB/FIM, NBP). Continuous, metric-bound.
- **Graceful degradation to topological/historical/ghost + discrete class + peer-validated calib + UNKNOWN-as-state in a BLE emergency mesh: NOT FOUND in one broad confirmatory search.** → the gap.

9. Research Questions, Layer, and the 10 Experiments That Would Move This From Design to Demonstrated

(Full experiment specs in GAPS.md; condensed falsification-relevant list here.)

#	Experiment	Serves	Pass criteria (example)
E1	RSSI log-distance calibration sweep	C4, RQ1	$\leq$ tier-1 σ on $\geq 70\%$ of devices
E2	Bearing: torso-scan vs two-walk	C5	median $\leq$ target bearing error
E3	Compass drift across walks	C4/C5	within hop tolerance
E4	BLE budget fit at 30-phone crowd	C3	PRR $\geq$ target, budget intact
E5	Multihop chain error accumulation	C6/C14	error within tier bounds ≤ 2 hops
E6	Ghost-decay navigation utility	C8	ghost beats unknown in success
E7	Triangle cycle-consistency detect rate	C10	detects $\geq X\%$ injected corrupt edges
E8	ANCHOR/DRIFTER stability under motion	C12	anchor frame stable vs drifter drift
E9	Mode-transition event integrity	C9	no silent mode flips
E10	Real 2-phone end-to-end demo	C1–C14	searcher reaches victim, phone-only

E10 is the gate. Until a 2-phone + relays real-device demo runs, claims are design-level only.

10. Deliverables Map

- **This document (RESEARCH_DUMP.md)** — clean synthesis, the "read this first" research document.
- **PAPER.md** — full paper draft (the companion).
- **SOURCES.md** — every source, with [VERIFIED]/[PARTIAL]/[UNVERIFIED] status, no fabrication.
- **GAPS.md** — updated gap registry, now distinguishing [RES] (design) from [EXP:NOT-RUN] (unexecuted) from [SPEC] (speculative).
- **website/** — static research + interview site (research/interview layering, prior-art, gap analysis, proposed system, interview synthesis).
- **README.md** — navigation.
- **Render note:** all deliverables rendered to PDF (A4) and mirrored to Downloads for the interview/PDF workflow.

End of RESEARCH_DUMP.md. Written under the project's honesty constitution: no fabricated citations, no fabricated results, no claimed prior art without verification. </content>